Markov Decision Processes: Foundations, Structure, and Importance

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1 Introduction to Markov Decision Processes

Markov Decision Processes (MDPs) provide a mathematical framework for modeling sequential decision-making under uncertainty. They formalize the interaction between an intelligent agent and its environment.

The origins of MDPs trace back to Andrey Markov (early 1900s), who introduced Markov chains to model stochastic processes with memoryless transitions. Later, Richard Bellman (1950s) developed dynamic programming and the principle of optimality, laying the foundation for optimal control theory. These ideas eventually unified into the modern theory of MDPs, now central to reinforcement learning, control systems, robotics, operations research, and artificial intelligence.

An MDP models how decisions affect future states probabilistically and how cumulative rewards determine optimal behavior.

An MDP is defined by the tuple:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$$

where:

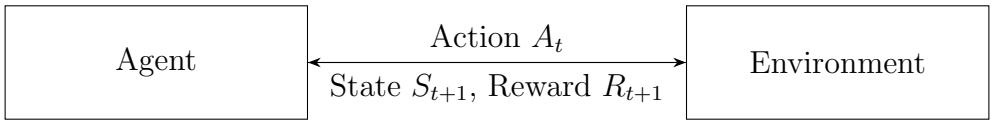
- $\mathcal{S}$: set of states
- $\mathcal{A}$: set of actions
- $P(s'|s, a)$: transition probabilities
- $R(s, a, s')$: reward function
- $\gamma \in [0, 1]$: discount factor

2 Agent–Environment Interaction

The agent interacts with the environment in discrete time steps:

$$S_t \to A_t \to R_{t+1}, S_{t+1}$$

2.1 Schematic Diagram



At each time step:

- Agent observes state S_t
- Agent selects action A_t
- Environment transitions to S_{t+1} according to $P(s'|s, a)$
- Agent receives reward R_{t+1}

3 Mathematical Foundations

3.1 Probability Theory

State transitions are governed by conditional probabilities:

$$P(S_{t+1} = s' \mid S_t = s, A_t = a)$$

For each state-action pair:

$$\sum_{s'} P(s'|s, a) = 1$$

3.2 Markov Property

The process satisfies:

$$P(S_{t+1}|S_t, S_{t-1}, \dots) = P(S_{t+1}|S_t)$$

Thus, the present state contains all necessary predictive information.

3.3 Discounting

Future rewards are geometrically discounted:

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

Discounting ensures:

- Convergence of infinite sums
- Preference for immediate rewards
- Mathematical contraction properties

3.4 Value Functions

State Value Function:

$$v_\pi(s) = \mathbb{E}_\pi[G_t|S_t = s]$$

Action Value Function:

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t|S_t = s, A_t = a]$$

3.5 Bellman Expectation Equation

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) \left[R(s, a, s') + \gamma v_\pi(s')\right]$$

3.6 Bellman Optimality Equation

$$v_*(s) = \max_a \sum_{s'} P(s'|s,a)\left[R(s,a,s') + \gamma v_*(s')\right]$$

The Bellman operator is a contraction mapping:

$$\|Tv - Tw\|_\infty \leq \gamma \|v - w\|_\infty$$

ensuring convergence to a unique optimal value function.

4 Example: Flow of a Markov Chain

Consider a simple Markov chain (no actions):

States:

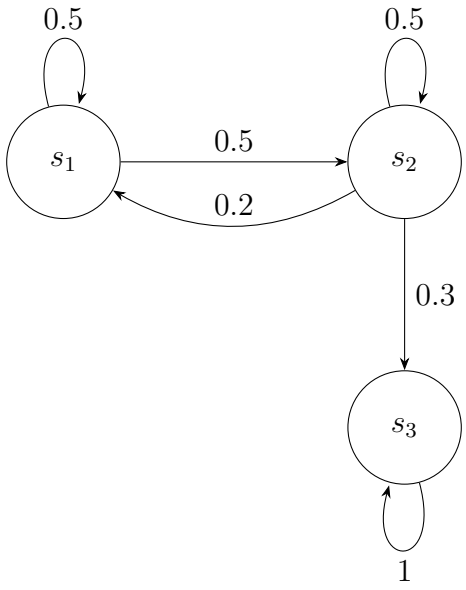
$$\mathcal{S} = \{s_1, s_2, s_3\}$$

Transition matrix:

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 \\ 0.2 & 0.5 & 0.3 \\ 0 & 0 & 1 \end{bmatrix}$$

Here, s_3 is an absorbing state.

4.1 State Transition Diagram



This illustrates how probability mass flows over time toward the absorbing state.

5 Reward Hypothesis and Importance of MDPs

The reward hypothesis states:

> All goals and purposes can be expressed as the maximization of expected cumulative reward.

This hypothesis unifies intelligence under a single optimization principle:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t R_{t+1}\right]$$

Why MDPs are important:

- Provide rigorous mathematical foundation for sequential decision-making
- Enable dynamic programming solutions
- Form basis of reinforcement learning algorithms
- Bridge probability theory, control theory, and optimization
- Allow provable convergence guarantees

MDPs transform decision-making from heuristic reasoning into structured mathematical optimization.

6 Conclusion

Markov Decision Processes unify stochastic dynamics, reward-driven objectives, and optimal control into a coherent mathematical framework. Through probability theory, the Markov property, value functions, and Bellman equations, MDPs enable principled reasoning about long-term consequences of actions.

They remain one of the foundational pillars of modern artificial intelligence and reinforcement learning.